Type MS · Simultaneous equations with powers and logs

Answer 1 of 6 · Question 9 part whole · 8 marks

2023 JAN · Paper 01 · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Simultaneous equations with powers and logs

Answer 2 of 6 · Question 10 part b · 9 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

10(a)	$\frac{1}{243}=3^{-5}$	B1
	$9^{3y}=3^{6y}$	B1
	$\frac{9^{3y}}{243}=3^{-5}\times3^{6y}\Rightarrow\frac{9^{3y}}{243}=3^{(6y-5)}$ *	M1A1 cso
		[4]
(b)	$27^{(x-2)}=3^{3(x-2)}=3^{(3x-6)}$	M1
	$6y-5=3x-6\Rightarrow(6y-3x-1=0)$	M1
	$\log_4 2=\frac{1}{2}$	B1
	$\log_{10}\sqrt{6xy}=\frac{1}{2}\log_{10}(6xy)\Rightarrow\log_{10}(6xy)=1$	M1
	$1=\log_{10}10\Rightarrow\log_{10}(6xy)=\log_{10}10\Rightarrow6xy=10$	M1
	$3x-6y-1=0$ $6xy=10$	
	Method A	
	$6y=\frac{10}{x}\Rightarrow3x-\frac{10}{x}-1=0\Rightarrow3x^2-x-10=0$	M1
	$3x^2-x-10=(3x+5)(x-2)=0\Rightarrow x=2,\ -\frac{5}{3}$	M1
	$3\times2-6y-1=0\Rightarrow6y=5\Rightarrow y=\frac{5}{6}$	A1
	$3\times\left(-\frac{5}{3}\right)-6y-1=0\Rightarrow-6y=6\Rightarrow y=-1$	
	$x=2\quad y=\frac{5}{6}$ or $x=-\frac{5}{3}\quad y=-1$	A1
		[9]
	Method B	
	$3x=\frac{5}{y}\Rightarrow\frac{5}{y}-6y-1=0\Rightarrow6y^2+y-5=0$	M1
	$6y^2+y-5=(6y-5)(y+1)=0\Rightarrow y=\frac{5}{6},\ -1$	M1
	$3x-6\times\frac{5}{6}-1=0\Rightarrow3x=6\Rightarrow x=2$	A1
	$3x-6\times(-1)-1=0\Rightarrow3x=-5\Rightarrow x=-\frac{5}{3}$	
	$x=2\quad y=\frac{5}{6}$ or $x=-\frac{5}{3}\quad y=-1$	A1
		[9]
		Total 13 marks

Type MS · Simultaneous equations with powers and logs

Answer 2 of 6 · Question 10 part b · 9 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
10	$\frac{9^{3y}}{243} = 27^{(x-2)}$ $\log_{10}\sqrt{6xy} = \log_4 2$	
(a)	Writes down $\frac{1}{243} = 3^{-5}$ May be seen as $\frac{1}{243} = \frac{1}{3^5}$ later correctly used as 3^{-5} when combining terms.	B1
	Writes down $9^{3y} = 3^{6y}$	B1
	Combines the terms $\frac{9^{3y}}{243} = 3^{-5} \times 3^{6y} = 3^{(6y-5)}$	M1
	For the correct expression with no errors. $\frac{9^{3y}}{243} = 3^{(6y-5)}$ *	A1 cso [4]
(b)	For dealing with the power of 3 to give $27^{(x-2)} = 3^{3(x-2)} = 3^{(3x-6)}$	M1
	For equating the powers of 3 to give the equation $6y-5 = 3x-6 \Rightarrow (6y-3x-1=0)$	M1
	For stating $\log_4 2 = \frac{1}{2}$	B1
	For dealing with the square root $\log_{10}\sqrt{6xy} = \frac{1}{2}\log_{10}(6xy) = \left(\frac{1}{2}\right) \Rightarrow \log_{10}(6xy) = 1$	M1
	For correctly removing all logarithms from the second equation $1 = \log_{10} 10 \Rightarrow \log_{10}(6xy) = \log_{10} 10 \Rightarrow 6xy = 10$	M1
	$3x - 6y - 1 = 0$ $6xy = 10$	
	Method A	
	For substituting $6y = \frac{10}{x}$ into the linear equation to give $3x - \frac{10}{x} - 1 = 0$ and attempting to form a 3TQ $3x^2 - x - 10 = 0$ OR for substituting $y = \frac{1}{6}(3x-1)$ into $6xy = 10$ to give $x(3x-1) = 10$ and attempting to form a 3TQ $3x^2 - x - 10 = 0$	M1
	For attempting to solve their 3TQ $3x^2 - x - 10 = (3x+5)(x-2) = 0 \Rightarrow x = 2, \ -\frac{5}{3}$	M1

Type MS · Simultaneous equations with powers and logs

Answer 2 of 6 · Question 10 part b · 9 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

	For finding the values of y $3\times2-6y-1=0\Rightarrow 6y=5\Rightarrow y=\frac{5}{6}$ $3\times\left(-\frac{5}{3}\right)-6y-1=0\Rightarrow -6y=6\Rightarrow y=-1$ $x=2 \quad y=\frac{5}{6}$ or $x=-\frac{5}{3} \quad y=-1$ Answers must be given in pairs, pairing may be implied from working.	A1 A1 [9]
	Method B	
	For substituting $3x=\frac{10}{2y}\Rightarrow 3x=\frac{5}{y}$ into the linear equation to give $\frac{5}{y}-6y-1=0$ and attempting to form a 3TQ $6y^2+y-5=0$ OR for substituting $x=\frac{1}{3}(6y+1)$ into $6xy=10$ to give $2(6y+1)y=10$ and attempting to form a 3TQ $6y^2+y-5=0$	M1
	For attempting to solve their 3TQ $6y^2+y-5=(6y-5)(y+1)=0\Rightarrow y=\frac{5}{6},\ -1$	M1
	For finding the values of x $3x-6\times\frac{5}{6}-1=0\Rightarrow 3x=6\Rightarrow x=2$ $3x-6\times(-1)-1=0\Rightarrow 3x=-5\Rightarrow x=-\frac{5}{3}$ $x=2 \quad y=\frac{5}{6}$ or $x=-\frac{5}{3} \quad y=-1$	A1 A1 [9]
		Total 13 marks

Type MS · Simultaneous equations with powers and logs

Answer 2 of 6 · Question 10 part b · 9 marks

2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Simultaneous equations with powers and logs

Answer 3 of 6 · Question 8 part whole · 8 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Number		
8	$\log_4 a + 2\log_4 b = \frac{5}{2}$	M1
	$\log_4(ab^2) = \frac{5}{2}$	M1
	$32 = ab^2$	A1
	$2^a = \frac{2^{16}}{2^{2b^2}}$	M1
	$a = 16 - 2b^2$ or $b^2 = 8 - \frac{1}{2}a$	A1
	$32 = a(8 - \frac{1}{2}a)$ or $32 = (16 - 2b^2)b^2$	M1
	$a^2 - 16a + 64 = 0$ or $2b^4 - 16b^2 + 32 = 0$	A1
	$a = 8$ $b = 2$	A1
		Total 8 marks

Type MS · Simultaneous equations with powers and logs

Answer 3 of 6 · Question 8 part whole · 8 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Guidance
Log equation	**Method 1 – Works in base 4**
M1	For an attempt to change the base of $3\log_8 b$ to base 4 using $\log_a x = \frac{\log_b x}{\log_b a}$ $3\log_8 b = \frac{3\log_4 b}{\log_4 8} = \frac{3\log_4 b}{\frac{3}{2}} = 2\log_4 b$ [accept $p\log_4 b$ where $p \neq 3$]
M1	Uses $n\log A = \log A^n$ **and** $\log A + \log B = \log AB$ to combine the logs correctly $\log_4(ab^2) = \frac{5}{2}$ [ft their p provided $p \neq 1$]
A1	For removing the logs in the equation to obtain $32 = ab^2$ o.e. e.g. $a^2b^4 = 1024$
	Method 2 – Works in base 8
M1	For an attempt to change the base of $\log_4 a$ to base 8 using $\log_a x = \frac{\log_b x}{\log_b a}$ $\log_4 a = \frac{\log_8 a}{\frac{2}{3}} = \frac{3\log_8 a}{2}$ [accept $q\log_8 a$ where $q \neq 1$]
M1	Uses $n\log A = \log A^n$ **and** $\log A + \log B = \log AB$ correctly to combine the logs $\log_8(a^{\frac{3}{2}}b^3) = \frac{5}{2}$ [ft their q]
A1	For removing the logs in the equation to obtain $a^{\frac{3}{2}}b^3 = 8^{\frac{5}{2}}$ **and** rearranges (raises both sides to the power of $\frac{2}{3}$) to obtain $32 = ab^2$
Second equation	
M1	For attempting to change the second equation to powers of 2 or 4: $2^a = \frac{2^{16}}{2^{2b^2}} \Rightarrow \left[2^a = 2^{(16-2b^2)}\right]$ or $4^{\frac{a}{2}} = \frac{4^8}{4^{b^2}} = \left(4^{\frac{a}{2}} = 4^{8-b^2}\right)$ At least one correct change of term e.g either 2^{16} or 2^{2b^2} OR either $4^{\frac{a}{2}}$ or 4^8
A1	Combines the powers to achieve $a = 16 - 2b^2$ or $\frac{a}{2} = 8 - b^2$ oe
Attempt to solve the simultaneous equations	
M1	For an attempt to solve their equations simultaneously, both of which must be in terms of a and b^2, to obtain a 3TQ in either a or b^2. $32 = a(8 - \frac{1}{2}a) \Rightarrow a^2 - 16a + 64 = 0$ or $32 = (16 - 2b^2)b^2 \Rightarrow 2b^4 - 16b^2 + 32 = 0$
M1	For an attempt to solve their 3TQ in either a or b^2 by any method. See General Guidance for the definition of an attempt For example: $a^2 - 16a + 64 = 0 \Rightarrow (a-8)(a-8) = 0 \Rightarrow a = ...$ $2b^4 - 16b^2 + 32 = 0 \Rightarrow b^4 - 8b^2 + 16 = 0 \Rightarrow (b^2-4)(b^2-4) = 0 \Rightarrow b = ...$
A1	For $a = 8$ and $b = 2$ [If $b = \pm 2$ is given as the final answer, withhold this mark].

Type MS · Simultaneous equations with powers and logs

Answer 3 of 6 · Question 8 part whole · 8 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Simultaneous equations with powers and logs

Answer 4 of 6 · Question 8 part whole · 8 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

8	$3\log_{27} x - \dfrac{\log_{27} y}{\log_{27} 3} = 2 \left[\Rightarrow 3\log_{27} x - 3\log_{27} y = 2\right]$ or $3\dfrac{\log_3 x}{\log_3 27} - \log_3 y = 2 \left[\Rightarrow \log_3 x - \log y = 2\right]$	M1M1
	$\log_{27}\left(\dfrac{x}{y}\right) = \dfrac{2}{3}$ or $\log_3\left(\dfrac{x}{y}\right) = 2$	M1
	$\Rightarrow \dfrac{x}{y} = 27^{\frac{2}{3}}$ or $\dfrac{x}{y} = 3^2 \left[\Rightarrow \dfrac{x}{y} = 9\right]$ or $\left[\log_6(x+3y) = 3\right] \Rightarrow x+3y = 6^3$	M1 A1
	$\Rightarrow x = 9y$ oe	
	$\Rightarrow 9y + 3y = 216$ oe	M1
	$\Rightarrow y = 18$	A1
	$\Rightarrow x\left[= 9\times 18\right] = 162$	A1 [8]
ALT	We are seeing increased use of $\log_{a^b} c = \dfrac{1}{b}\log_a c$ During marking, this can be considered as 'a change of base of log' and in this question leads to the following solution. The notes can still be applied. If using this method, we must see a full change of base ie $\left[\log_{27} x = \log_{3^3} x =\right] \dfrac{1}{3}\log_3 x$ or $\left[3\log_{27} x = 3\log_{3^3} x =\right] \log_3 x$ $\left[\log_{27} x^3 = \log_{3^3} x^3 =\right] \dfrac{1}{3}\log_3 x^3$ or combining 1st 2 M marks with no errors $\log_{27} x^3 = \dfrac{3\log_3 x}{3}$ Leaving as $3\log_{3^3} x$ or $\log_{3^3} x^3$ is insufficient	
	$\left[3\log_{3^3} x - \log_3 y\right] = 2 \Rightarrow \dfrac{3\log_3 x}{3} - \log_3 y = 2 \left[\Rightarrow \log_3 x - \log_3 y = 2\right]$	M1M1
	$\log_3\left(\dfrac{x}{y}\right) = 2$	M1
	$\Rightarrow \dfrac{x}{y} = 3^2$ or $\left[\log_6(x+3y) = 3\right] \Rightarrow x+3y = 6^3$	A1
	Final M1 A1 A1 as main scheme.	
		Total 8 marks

Type MS · Simultaneous equations with powers and logs

Answer 4 of 6 · Question 8 part whole · 8 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
M1	For any correct use of the power rule for logs – this mark may be awarded at any point in working for **any** correct change use of the power rule.
M1	For **any correct change of base of log** – this mark may be awarded at any point in working for **any** correct change of base of log.
M1	For applying the division law correctly – this mark may be awarded at any point in working for **any** correct application of the division law.
M1	For correctly converting a log equation into an exponential – – this mark may be awarded at any point in working for **any** correct conversion of a log equation into an exponential equation (particularly watch out for this also when the candidate is manipulating the second equation).
A1	For obtaining $x = 9y$ oe
M1	For solving a pair of **linear** simultaneous equations in x and y. Allow one error. There must be an attempt at a method. Candidates must have gained at least any 2 of the previous 4 method marks before they can be awarded this mark. Correct final answers would imply this method mark.
A1	For the correct value of x or y Candidates are told they must use an algebraic method, so we must see the work for all the previous marks for this to be awarded. Trial and error will score 0.
A1	For both correct values. Candidates are told they must use an algebraic method, so we must see the work for all the previous marks for this to be awarded. Trial and error will score 0.
If different approaches are used – apply the mark scheme and award marks accordingly.	

Type MS · Simultaneous equations with powers and logs

Answer 4 of 6 · Question 8 part whole · 8 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Simultaneous equations with powers and logs

Answer 5 of 6 · Question 9 part whole · 7 marks

2020 NOV · Paper 01 (regional R) · find part whole in this mark scheme extract

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Type MS · Simultaneous equations with powers and logs

Answer 6 of 6 · Question 11 part whole · 8 marks

2023 NOV · Paper 01 · extract 1 of 6 · find part whole in this mark scheme extract

11 **Eg1**	$2\log_4 x = \log_3 3y^2$ $\frac{2\log_2 x}{\log_2 4} = \log_3 3 + \log_3 y^2 \Rightarrow \frac{2\log_2 x}{\log_2 4} - 2\log_3 y = 1$ $\Rightarrow \log_2 x - 2\log_3 y = 1$ oe $\log_2 x^3 + 8\log_9 y = 13$ $3\log_2 x + \frac{8\log_3 y}{\log_3 9} = 13 \Rightarrow 3\log_2 x + 4\log_3 y = 13$ oe Solves the simultaneous equations by any method to find a value for $\log_2 x$ or $\log_3 y$ $3\log_2 x + 4\log_3 y = 13$ oe $\log_2 x - 2\log_3 y = 1$ oe $\Rightarrow \log_2 x = \frac{15}{5}$ or $\log_3 y = \frac{10}{10}$ oe $x = 2^{\frac{15}{5}} = 8$ $y = 3^{\frac{10}{10}} = 3$	M1B1 M1M1 dddM1 A1 M1 A1 [8]
Eg2	$2\log_4 x = \log_3 3y^2$ $2\log_4 x = \log_3 3 + \log_3 y^2 \Rightarrow 2\log_4 x = 1 + 2\log_3 y$ oe $\log_2 x^3 + 8\log_9 y = 13$ $\frac{\log_4 x^3}{\log_4 2} + \frac{8\log_3 y}{\log_3 9} = 13 \Rightarrow 2\log_4 x^3 + 4\log_3 y = 13$ $\Rightarrow 6\log_4 x + 4\log_3 y = 13$ oe Solves the simultaneous equations by any method to find a value for $\log_4 x$ or $\log_3 y$ $2\log_4 x = 1 + 2\log_3 y$ oe $6\log_4 x + 4\log_3 y = 13$ oe $\Rightarrow \log_4 x = \frac{15}{10}$, or $\log_3 y = \frac{10}{10}$ oe $x = 4^{\frac{15}{10}} = 8$ $y = 3^{\frac{10}{10}} = 3$	M1B1 M1M1 dddM1 A1 M1 A1 [8]

Type MS · Simultaneous equations with powers and logs

Answer 6 of 6 · Question 11 part whole · 8 marks

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Eg3	$2\log_4 x = \log_3 3y^2$ $2\log_4 x = \dfrac{\log_9 3y^2}{\log_9 3} \Rightarrow \log_4 x = \log_9 3y^2 \Rightarrow \log_4 x = \log_9 3 + \log_9 y^2$ $\Rightarrow \log_4 x = 0.5 + 2\log_9 y$ oe $\log_2 x^3 + 8\log_9 y = 13$ $\dfrac{\log_4 x^3}{\log_4 2} + 8\log_9 y = 13 \Rightarrow \log_4 x^3 + 4\log_9 y = \dfrac{13}{2}$ $\Rightarrow 3\log_4 x + 4\log_9 y = \dfrac{13}{2}$ oe Solves the simultaneous equations by any method to find a value for $\log_4 x$ or $\log_9 y$ $3\log_4 x + 4\log_9 y = \dfrac{13}{2}$ oe $\log_4 x = 0.5 + 2\log_9 y$ oe $\Rightarrow \log_4 x = \dfrac{15}{10}$ or $\log_9 y = \dfrac{5}{10}$ oe $x = 4^{\frac{15}{10}} = 8 \qquad y = 9^{\frac{5}{10}} = 3$	M1B1 M1M1 dddM1 A1 M1 A1 [8]
An aid to marking (please also see more detailed notes).		
Correct application of addition/subtraction law, anywhere in working. Once seen, ignore any other incorrect applications.		M1
Any correct evaluation of a $\log_a a^b = b$ seen. Look for implicit evaluation during change of base of log. a, b rational.		B1
Correct application of the power law seen once anywhere. Once seen, ignore any other incorrect applications.		M1
Any correct change of base seen once anywhere. Once seen, ignore any other incorrect applications.		M1
Solve a pair of simultaneous equations – one error permitted, see detailed notes for required form.		dddM1
One correct log equation following from this solution.		A1
Undoing a log of the required form – see detailed notes.		M1
Both correct values of x and y.		A1
Note also that: $2\log_4 x = \log_2 x$ **or** $8\log_9 y = 4\log_3 y$ **or** $\log_2 x^3 = 2\log_4 x^3$ **or** $\log_3 3y^2 = 2\log_9 3y^2$ will normally imply B1 and 3rd M1		

Type MS · Simultaneous equations with powers and logs

Answer 6 of 6 · Question 11 part whole · 8 marks

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ALT1	$2\log_4 x = \log_3 3y^2$	
	$\frac{2\log_2 x}{\log_2 4} = \log_3 3y^2 \Rightarrow \log_2 x = \log_3 3y^2$	M1
	$\log_2 x^3 + 8\log_9 y = 13 \Rightarrow 3\log_2 x + 8\log_9 y = 13$	B1 M1
	$3\log_3 3y^2 + 8\log_9 y = 13$	M1
	$\log_3 27y^6 + 8\frac{\log_3 y}{\log_3 9} = 13$	dddM1
	$\log_3 27y^6 + 4\log_3 y = 13$	
	$\log_3 27y^6 + \log_3 y^4 = 13$	
	$\log_3 27y^{10} = 13$	A1
	$27y^{10} = 3^{13}$	
	$y = \sqrt[10]{\frac{3^{13}}{27}} = 3$	M1
	$\log_2 x = \log_3 3\times 3^2 \Rightarrow \log_2 x = 3$	A1
	$x = 2^3 = 8$	[8]
ALT2	$2\log_4 x = \log_3 3y^2$	
	$3\log_4 x = \frac{3}{2}\log_3 3y^2 \Rightarrow \log_4 x^3 = \frac{3}{2}\log_3 3y^2$	M1
	$\frac{\log_2 x^3}{\log_4 2} = \frac{3}{2}\log_3 3y^2 \Rightarrow \log_4 x^3 = 3\log_3 3y^2$	B1 M1
	$3\log_3 3y^2 + 8\log_9 y = 13$	M1
	$\log_3 27y^6 + 8\frac{\log_3 y}{\log_3 9} = 13$	
	$\log_3 27y^6 + 4\log_3 y = 13$	
	$\log_3 27y^6 + \log_3 y^4 = 13$	dddM1
	$\log_3 27y^{10} = 13$	A1
	$27y^{10} = 3^{13}$	
	$y = \sqrt[10]{\frac{3^{13}}{27}} = 3$	M1
	$\log_2 x = \log_3 3\times 3^2 \Rightarrow \log_2 x = 3$	A1
	$x = 2^3 = 8$	[8]
		Total 8 marks

Type MS · Simultaneous equations with powers and logs

Answer 6 of 6 · Question 11 part whole · 8 marks

2023 NOV · Paper 01 · extract 4 of 6 · find part whole in this mark scheme extract

Mark	Notes
M1	For any correct application of the addition (or subtraction) law eg $\log_3 3y^2 = \log_3 3 + \log_3 y^2$ Seen anywhere in their working once, ignore any other incorrect applications.
B1	For $\log_3 3 = 1$ **or** if students do a change of base first, this mark can also be awarded for example for $\log_2 4 = 2$ or $\log_3 9 = 2$ or $\log_4 2 = 0.5$ or $\log_9 3 = 0.5$ - note that this evaluation may not be explicitly seen, but seen in the simplification of an equation, particularly following a change of base of logs. **This mark may be awarded for any correct use of $\log_a a^b = b$ seen. *a, b* rational.**
M1	For applying the power law correctly with **either** $\log_p x^3$ **or** $\log_q y^2$ $p = 2$ or 4 $q = 3$ or 9 Seen anywhere in their working once, ignore any other incorrect applications.
M1	For changing the base of any log in either equation. eg $2\log_4 x = \frac{2\log_2 x}{\log_2 4}$ **or** $8\log_9 y = \frac{8\log_3 y}{\log_3 9}$ **or** $\log_2 x^3 = \frac{\log_4 x^3}{\log_4 2}$ **or** $\log_3 3y^2 = \frac{\log_9 3y^2}{\log_9 3}$ **This mark may be awarded for any correct change of base of logs seen.** Seen anywhere in their working once, ignore any other incorrect applications.
dddM1	For solving their simultaneous equations by any valid complete method. Allow one error. Candidates must reach a value for $\log_p x^n$ or $\log_q y^m$ $p = 2$ or 4 $q = 3$ or 9 m, n constants Dependent on all 3 previous method marks. The equations must be in a form with $\log_p ax^n$ and $\log_q by^m$ in $p = 2$ or 4 $q = 3$ or 9 All methods of solving (substitution and elimination being the most common) are valid.
A1	For the correct values of $\log_p x^n$ or $\log_q y^m$ $p = 2$ or 4 $q = 3$ or 9 m, n constants Markers will need to check the solution of candidates' simultaneous equations, but more than one error seen will result in dddM0 and A0. **The question asks students to show their working clearly.** **This accuracy mark would generally only be gained after clearly seeing the work for the previous 4 method marks. If there is a solution where it is clear the student is working more concisely and fully correctly, perhaps taking shortcuts or missing the odd step – please send this to Review.**
M1	For a correct method to 'undo' the logs on any equation of the form $\log_p ax^n = E$ or $\log_q by^m = F$ $p = 2$ or 4 $q = 3$ or 9 , E, m and n are non-zero constants.
A1	For the correct values of x and y

Type MS · Simultaneous equations with powers and logs

Answer 6 of 6 · Question 11 part whole · 8 marks

2023 NOV · Paper 01 · extract 5 of 6 · find part whole in this mark scheme extract

ALTS	**For any similar methods mark using the same principles, but send to Review, if not clear how to implement these.**
ALT1 M1	For change of base of log eg $2\log_4 x = 2\dfrac{\log_2 x}{\log_2 4}$ or $8\log_9 y = 8\dfrac{\log_3 y}{\log_3 9}$ **This mark may be awarded for any correct change of base of logs seen.**
B1	For $\log_2 4 = 2$ or $\log_3 9 = 2$. **This mark may be awarded for any correct use of $\log_a a^b = b$ seen. *a, b* rational.**
M1	For applying the power law correctly with $\log_2 x^3$ or in reverse $3\log_3 3y^2 \Rightarrow \log_3 27y^6$
M1	For correctly using $\log_2 x = \log_3 3y^2$ in $3\log_2 x + 8\log_9 y = 13$
dddM1	For using the addition (or subtraction) law in reverse $\log_3 27y^6 + \log_3 y^4 \Rightarrow \log_3 27y^{10}$
A1	For $\log_3 27y^{10} = 13$
M1	For a correct method to undo logs.
A1	For the correct values of x and y
ALT2 M1	For applying the power law correctly in reverse $3\log_2 x = \log_2 x^3$
B1	For $\log_2 4 = 2$ or $\log_3 9 = 2$. **This mark may be awarded for any correct use of $\log_a a^b = b$ seen. *a, b* rational.**
M1	For change of base of log eg $2\log_4 x = 2\dfrac{\log_2 x}{\log_2 4}$ or $8\log_9 y = 8\dfrac{\log_3 y}{\log_3 9}$ **This mark may be awarded for any correct change of base of logs seen.**
M1	For correctly using $\log_4 x^3 = 3\log_3 3y^2$ in $3\log_3 3y^2 + 8\log_9 y = 13$
dddM1	For using the addition (or subtraction) law in reverse $\log_3 27y^6 + \log_3 y^4 \Rightarrow \log_3 27y^{10}$
A1	For $\log_3 27y^{10} = 13$
M1	For a correct method to undo logs.
A1	For the correct values of x and y

Type MS · Simultaneous equations with powers and logs

Answer 6 of 6 · Question 11 part whole · 8 marks

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